

Param Jain

Boston, MA · +1 (302) 409-9135 · jain.param@northeastern.edu · [Portfolio](#) · [linkedin.com/in/paramsachinjain](https://www.linkedin.com/in/paramsachinjain) · github.com/paramjain13

Computer Science master's student writing production software in Python, C++, and TypeScript, with a foundation in data structures, algorithms, and object-oriented design. Builds distributed data systems on Linux, automates routine engineering work through scripting, and instruments services with monitoring and alerting.

Education

M.S. in Computer Science (Data Science and AI Focus) | Northeastern University, Boston, MA

Relevant Coursework: Algorithms, Data Structures, Machine Learning, Artificial Intelligence, Database Management Systems

B.Tech in Computer Science | Medi-Caps University, Indore, India

Work Experience

Software Engineer Intern | DevQAExpert Solution Private Limited

May 2024 – Aug 2025

- Developed Python scripts that automated routine validation work across 15,000+ scenarios per release, ranking them by failure risk so the highest-risk cases ran first; improved defect prediction accuracy 42% and removed manual effort from every release cycle
- Built automation infrastructure integrating test execution into Jenkins and GitHub Actions CI/CD, querying result data through SQL and REST APIs rather than manual exports; reduced manual effort 60% and improved coverage consistency across engineering workflows
- Implemented monitoring and alerting across 12 deployment cycles tracking runtime, failures, and latency anomalies, then troubleshooted each at code level to isolate root cause; surfaced 40+ defects per release before software reached production
- Analyzed competing approaches to release validation and chose the one the team adopted across 12 deployment cycles, then supported peers and managers in running it; reduced code review turnaround 30% across scheduled releases

Machine Learning Intern | Genesis Technologies

Jan 2023 – Dec 2023

- Designed, tested, deployed, and maintained an end-to-end machine learning system in Python serving live production traffic; improved classification accuracy from 30% to 95% across 30K+ monthly users through iterative evaluation of competing approaches
- Engineered data processing in Python and SQL to ingest, clean, deduplicate, and validate large record volumes at scale; built 15+ features from image and document quality signals that separated valid inputs from faulty ones
- Deployed and maintained containerized backend services on AWS with Docker and Flask REST APIs, monitoring latency and error rates under live traffic; sustained 97% uptime across 1,000+ daily requests with alerting on degradation
- Automated evaluation comparing every candidate build against the previous release under identical conditions, analyzing results to pick the best solution; lifted F1 from 0.71 to 0.89 through threshold tuning and systematic error analysis

Data Analyst Intern | Agrawal's 420 Namkeen, Sweets & Bakery

Jan 2022 – Jun 2022

- Optimized large-scale Python and SQL data pipelines across Oracle and PostgreSQL, profiling slow query paths and rewriting join and scheduling logic; cut pipeline runtime 45% and guaranteed on-time delivery for downstream systems
- Wrote scheduled automation processing large volumes of operational records and validating completeness at each stage before downstream querying; improved freshness and reliability of the daily analysis consumed across multiple teams
- Built monitoring and automated alerting tracking job latency, failures, and data freshness across scheduled runs; surfaced degraded jobs immediately and accelerated root-cause investigation for the teams depending on them
- Created dashboards and written reports translating system-level metrics into findings other teams could act on, working directly with peers and managers to define what each metric needed to surface

Technical Skills

Python, C++, C, TypeScript, JavaScript, SQL, Bash, Rust, Data Structures, Algorithms, Object-Oriented Design, Distributed and Parallel Systems, Unix and Linux, Large Software Systems, Scripting and Automation, Monitoring and Alerting, Information Retrieval, Natural Language Processing, Machine Learning, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Kafka, PostgreSQL, MySQL, MongoDB, Apache Airflow, ETL Pipelines, Grafana, Web Application Development, React, Next.js, FastAPI, Flask, REST APIs, Docker, CI/CD Pipelines, GitHub Actions, Git, AWS (S3, Lambda, EC2)

Projects

Navigation Validation Simulator | C++ Simulation and Release Automation

Aug 2026 – Present

- Built a C++ simulator on Linux that replays navigation routes against expected outputs and gates releases at a 98% accuracy threshold, blocking regressions before deployment
- Automated the full validation sweep through GitHub Actions and Docker so every commit runs under identical conditions without manual setup, making results repeatable across builds
- Developed a root-cause module that parses failure logs and groups routes by failure signature, turning raw output into a traceable summary for debugging a blocked release

Fleet Telemetry Observability Platform | Distributed Pipeline with Monitoring

Aug 2026 – Present

- Built a distributed Kafka and PostgreSQL pipeline on Linux ingesting telemetry from 1,000+ simulated devices at 10,000+ events per second, validating each stream before storage
- Implemented 3-sigma anomaly detection over streaming data to flag traffic anomalies automatically rather than requiring manual inspection of raw logs
- Defined system health metrics and built Grafana dashboards surfacing latency, throughput, and error-rate trends, with documentation covering each metric and how to read it

NL2SQL Agent | Information Retrieval and Query Verification

Jun 2026 – Aug 2026

- Built a natural language to SQL system in Python reaching 64% execution accuracy across 500 benchmark questions against a 1M+ row database through iterative self-correction
- Designed a static verifier using AST parsing and foreign-key graph traversal that validates every generated query before execution, applying graph algorithms to catch invalid join paths
- Implemented cost-aware routing after profiling per-query cost and latency, evaluating tradeoffs to cut compute cost while holding execution accuracy steady across the benchmark

Publications & Certifications

- Published "Enhancing Human-Robot Interaction through Advanced NLP" in IJSREM (2024), integrating BERT and GPT-3 for real-time speech recognition and emotion detection; achieved 86.5% interaction accuracy at sub-500ms response latency
- AWS Academy coursework: Cloud Foundations, Data Engineering, and Generative AI Foundations (2026)